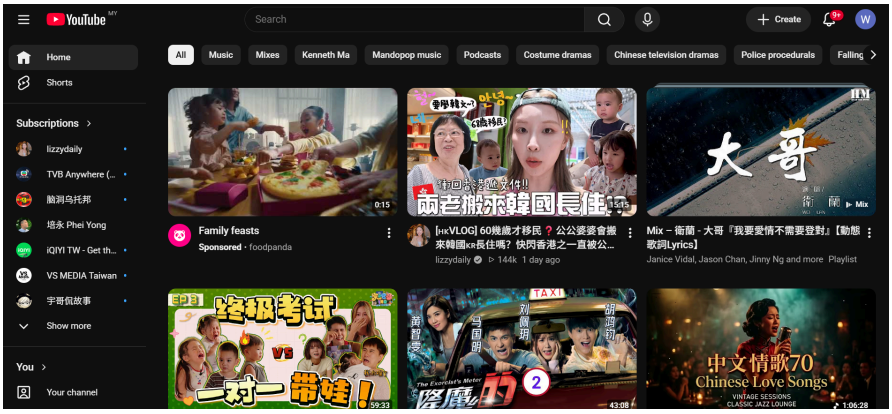
Topic 1_ Lab

By:

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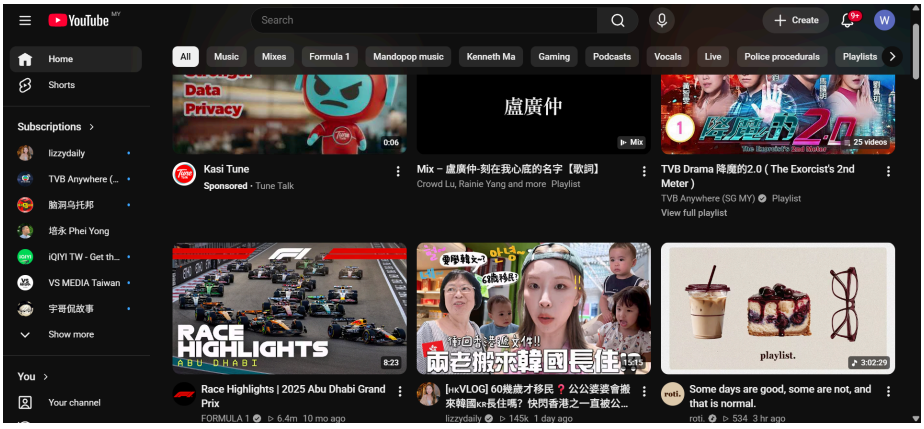
A1: Interact with assigned system

1. What is recommended on your home page right now? Can you explain why?



My YouTube home page displays a mixture of content, including Chinese television dramas, Chinese music mixes, family and parenting videos**.** This is because YouTube may use my previo us viewing history, search activity and subscriptions to personalise recommendations. The variety of content suggests that the system may be combining several interests rather than focusing on one topic.

2. Search for something unusual — does the recommendation feed change?



After I searched for Formula 1, my recommendation feed included a Formula 1 video while some videos from my original home page remained. The feed did not completely replace the previous recommendations.

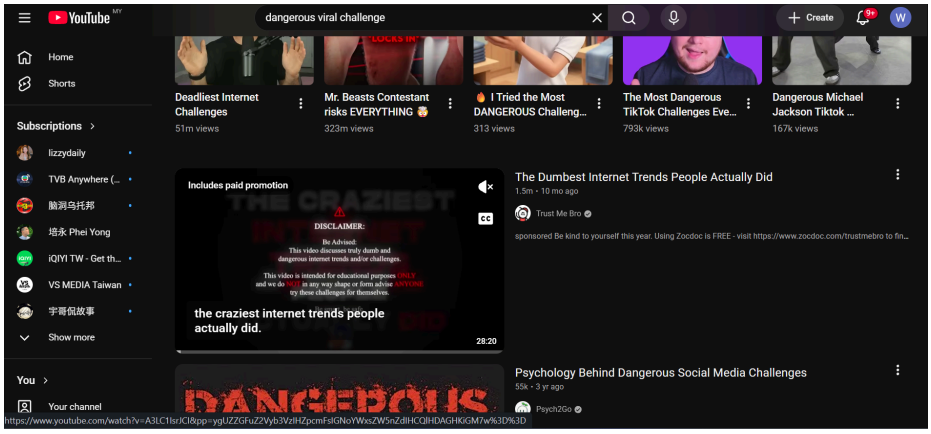
YouTube may incorporate recent search activity into its recommendations while continuing to use signals from my previous interests.

3. Can you observe differences across content types, or identify factors that might cause recommendations to differ across users? Distinguish clearly between what you observed and what you are hypothesising.

Observation: My home page contains several different content types, including music, vlogs, television dramas and sponsored content. After searching for Formula 1, a Formula 1 video appeared alongside some of the original recommendations.

Hypothesis: YouTube may use different signals for different content types. For example, music recommendations might be influenced by listening or viewing history, while sports recommendations might respond to recent searches and video engagement. Recommendations may also differ between users because they have different search histories, subscriptions, watch histories and viewing preferences.

4. Can you find any content that seems inappropriate or harmful being recommended?



After searching for “dangerous viral challenge”, I found videos about deadly internet challenges and dangerous TikTok trends. These topics involve potential safety risks, but some videos provide educational warnings rather than encourage harmful behaviour.

5. How transparent is the system about WHY it recommends something?

YouTube displays video titles, channel names and other information, but my home page does not explain why each video is recommended. Therefore, its transparency appears limited from the home page alone. I would need to check its recommendation explanations for more information.

A2: Complete the Observation Sheet

System name	Youtube or Spotify Recommendations
Group members	1. Mun Weng Yann A24AI0067 2. Faqihah Humaira' A24AI0028 3. Nur Nadsyuha A24AI0117 4. Saveetha Shree A24AI0086
1. What problem does this system solve? Who benefits?	Problem: Youtube was created to solve the difficulty of sharing video files online. The recommendation system solves this problem by using AI to personalize video suggestions based on users' interest and history. Beneficiaries: All users across ages
2. Which AI stack layer is most visible to users?	Application Development - Users see their own displays personalized video recommendations through the homepage.
3. Quality dimensions (at least 3)	Maintainability: Allows recommendation models to update based on users' search. Reliability: Consistently delivers relevant recommendations quickly when users open Youtube. Scalability: Able to handle millions of users and videos daily and simultaneously.
4. What did the system do WELL? Provide a specific example and evidence.	As mentioned in A1(2), YouTube recommended new content immediately after I searched for Formula 1. This shows that Youtube can quickly adapt its recommendations based on recent user activity while keeping some videos from my original homepage. Youtube's recommendation system balances newly detected interests with previous viewing preferences instead of completely replacing existing homepage.
5. What did the system do POORLY? Provide a specific example and	As mentioned in A1(4), YouTube still displayed videos related to dangerous viral challenges and harmful TikTok trends. This raises concerns about the system's safety, as users may still

evidence.	be exposed to potentially harmful content. The system may struggle to distinguish between content that warns against dangerous behaviour and content that promotes it.
6. Responsible AI risks identified (Kleppmann & Riccomini, 2026, Ch. 14)	Privacy and tracking: Youtube uses search history, watch history and engagement patterns to personalize recommendations.
7. If you could fix ONE thing	I would improve Youtube's recommendations algorithm to better analyze and understand video content, distinguishing between harmful and educational content and I would also encourage the system to recommend more videos from smaller creators.

B4: Reflection

Based on our conceptual system map, the Data Processing/Input Pipeline and the Safety Checks component had the most critical failure points

1. Data Processing pipeline had the risks of misinterpreting mixed viewing contents or treating a single "one-off" query as the user's permanent interest, which may expand the recommendations for sensitive or unwanted topics affecting the reliability and robustness
2. The safety checks layer had suffered from misclassification distinguishing between awareness/educational danger warning videos and content that actually promoted dangerous trends resulting in safety risks.

This reflects that AI engineering should also focus on the engineering system around the model rather than relying solely on the model's capabilities. They can focus on

1. Safety & Alignment Guardrails when deploying the input/output guardrails at the Application layer to filter the contents
2. Multi-Dimensional Quality Trade-Offs to optimise feature pipelines to understand short-term search signals to balance safety,fairness and relevance so one-off queries doesn't affect user feeds